Automotive Solutions

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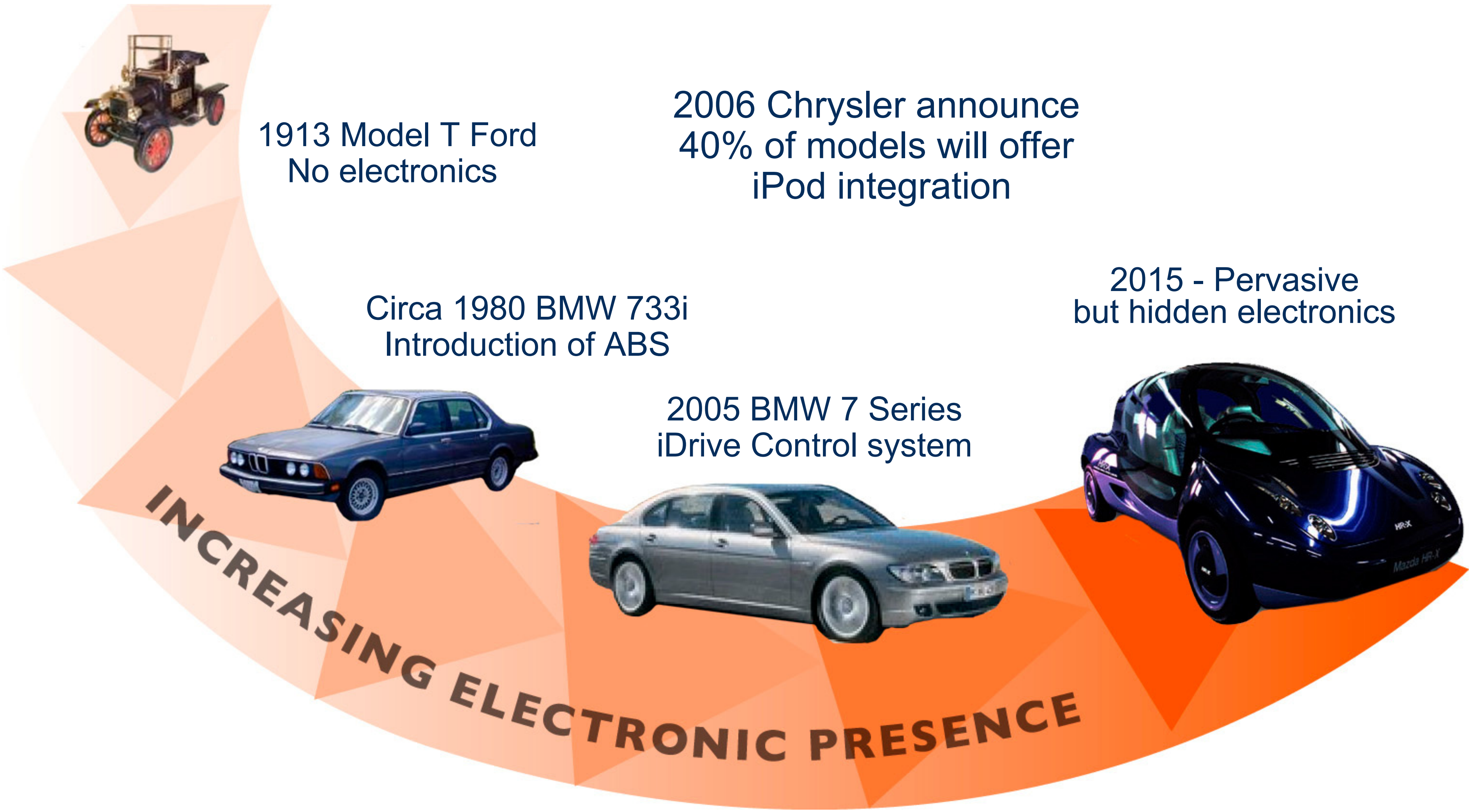
1906

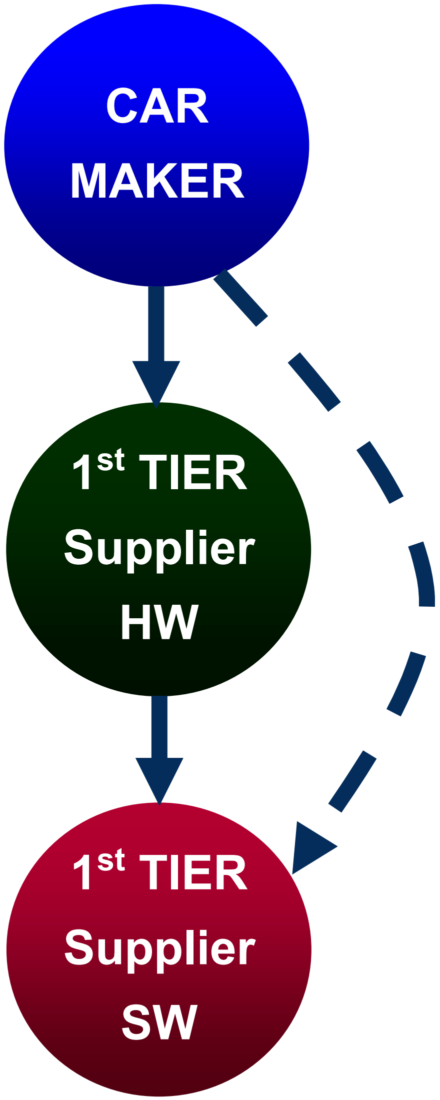
- First Grand Prix

2006 - Intelligence is Driving Change



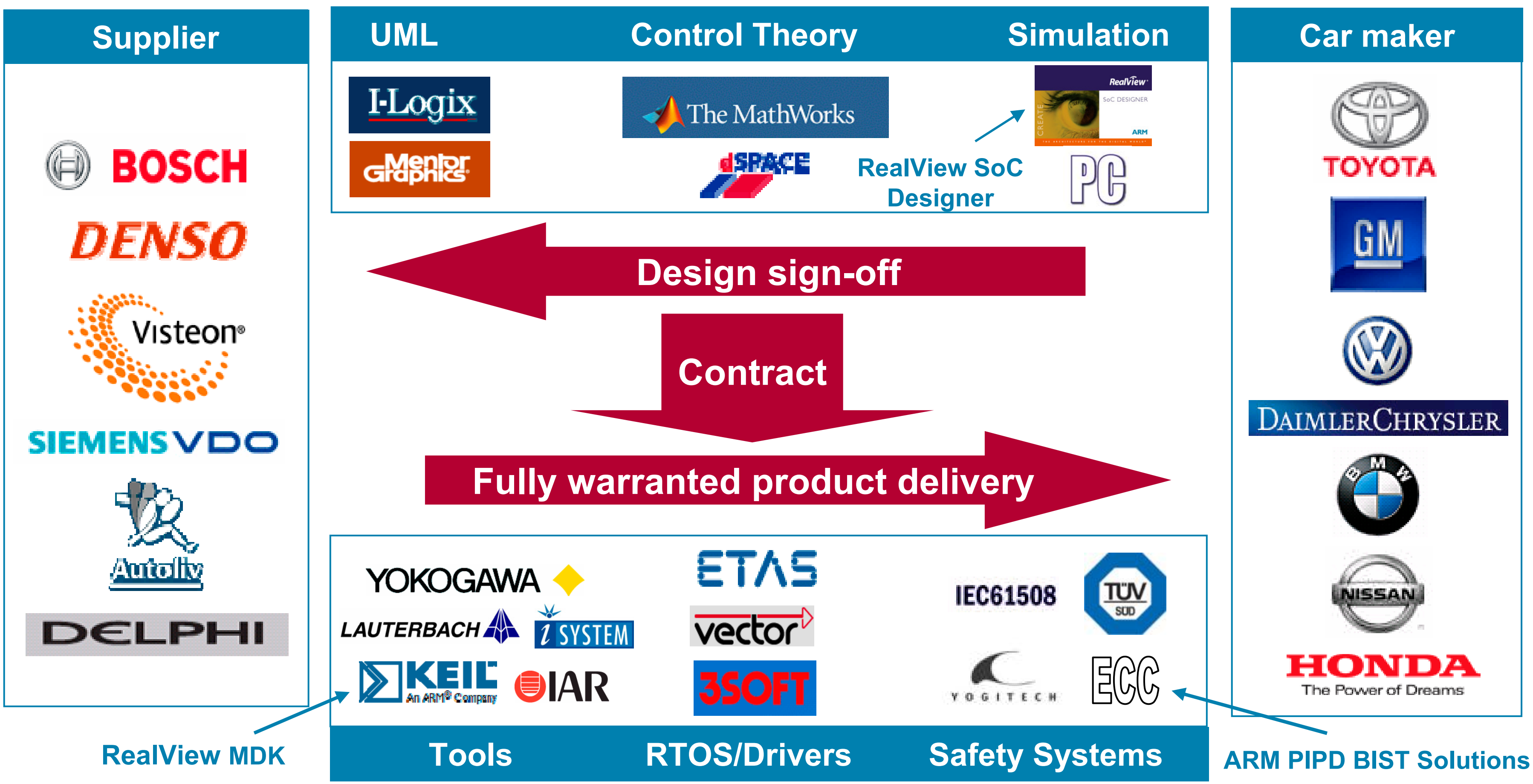
Software in Automotive

- Standard processor architecture allows
  - Software reuse
  - Compiler and tools standardisation
- Software cost increasing
  - 9% vehicle build cost today
  - 13% by 2010
- Supplier’s hardware modules mask the real development costs
  - Software by rolled into ‘piece’ prices
  - ‘Software is for free’
- Hidden costs limit efficient cost control

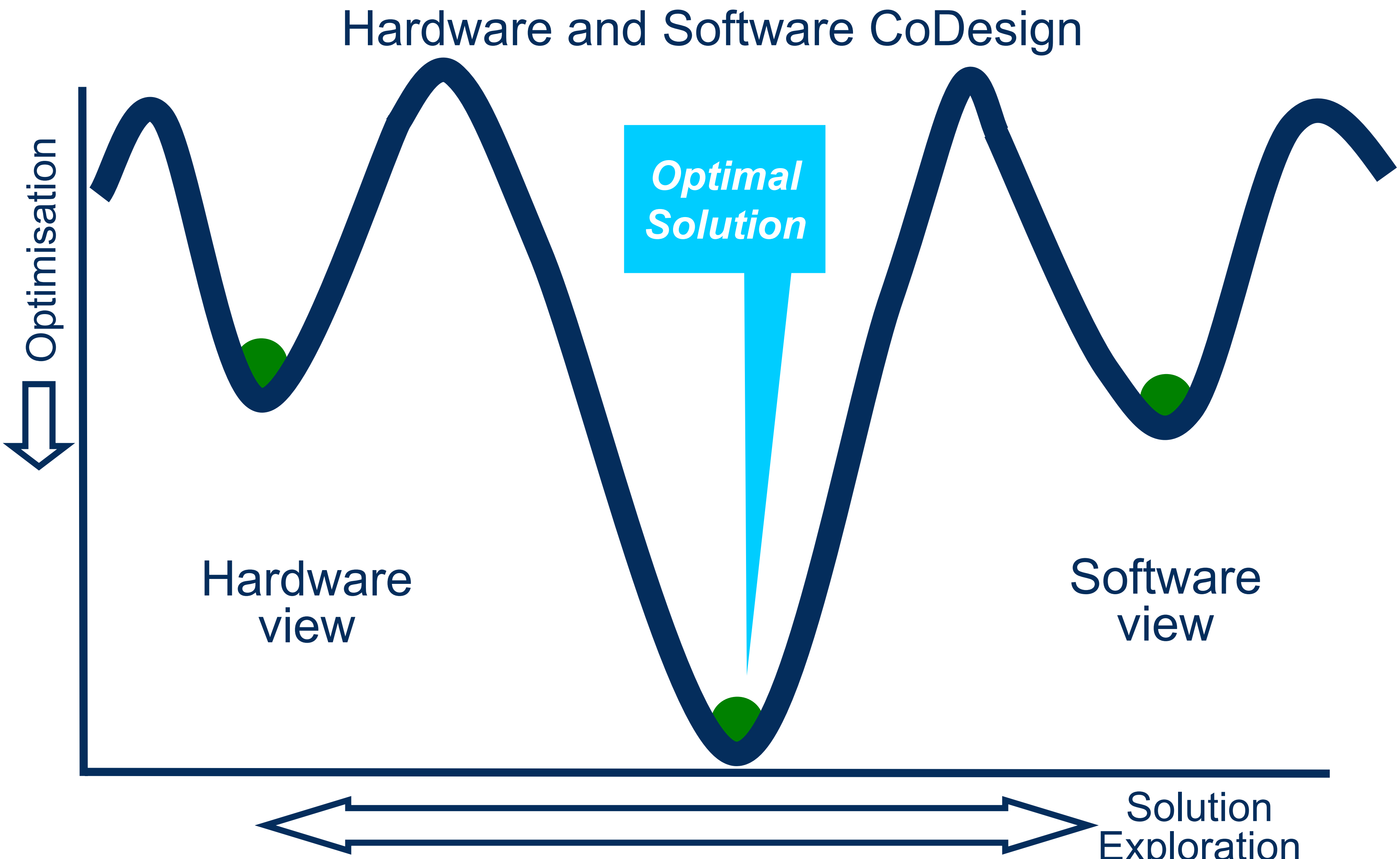


Automotive – Increased Modeling Use

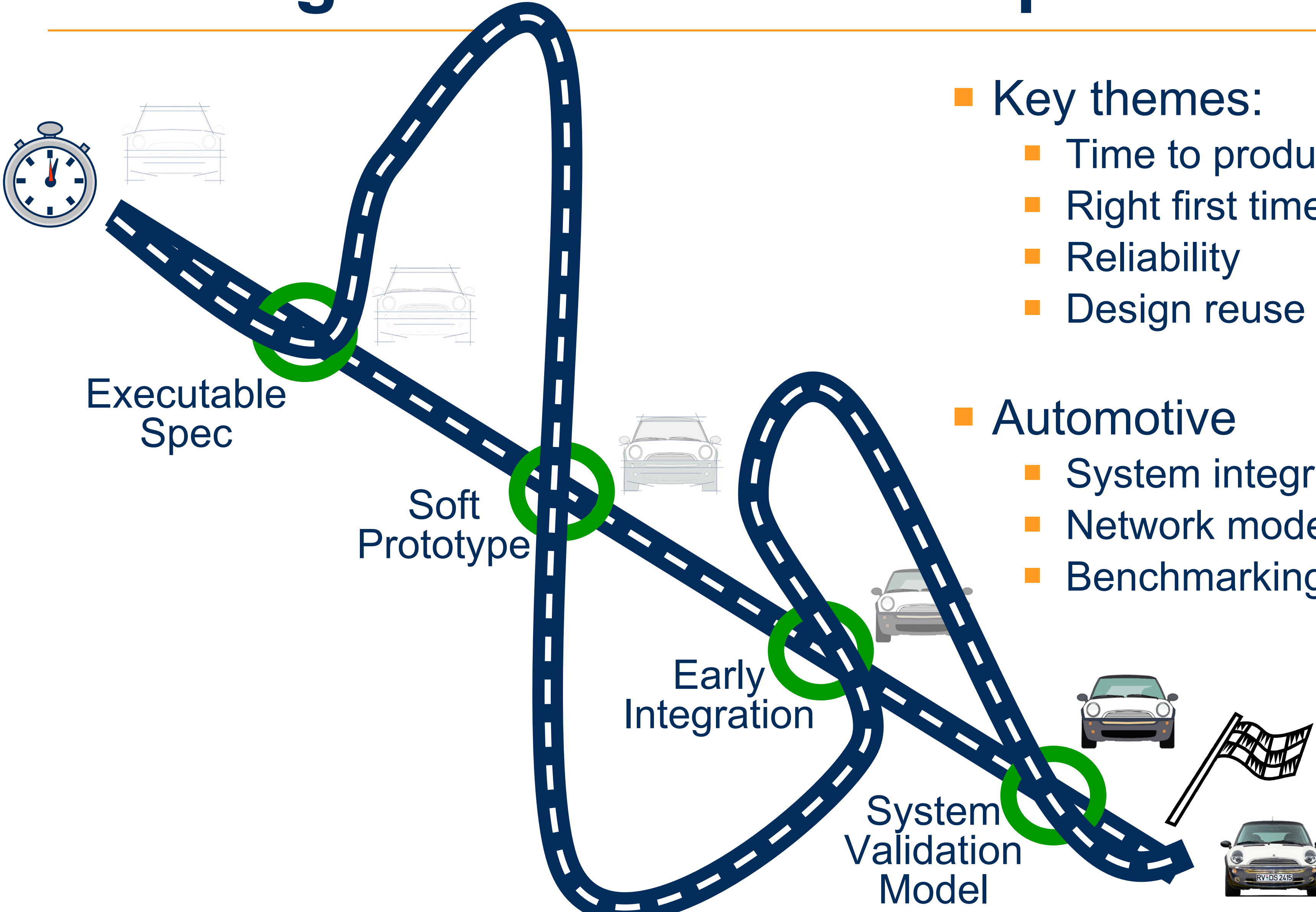
- Modeling growing in importance for product development and sign-off
- Estimates place it at up to $17M per Module Program



ESL Paradigm Shift in Design



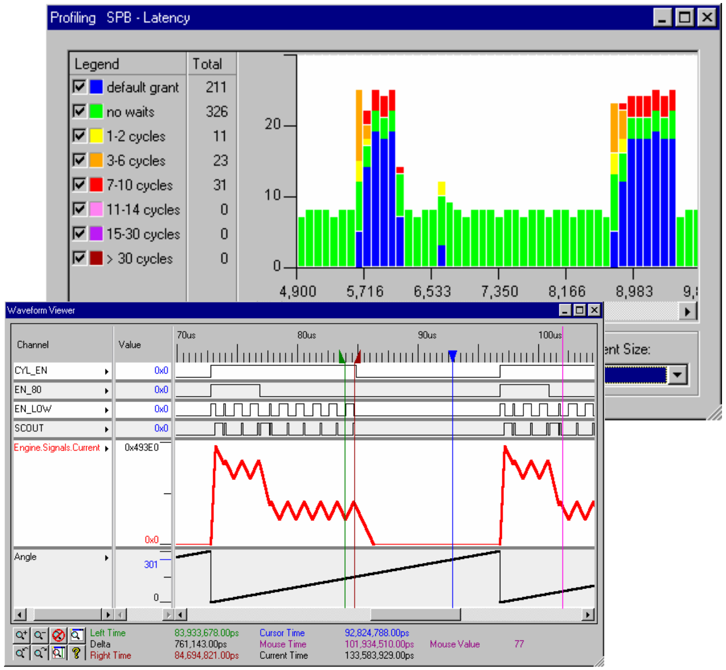
Entering Concurrent Development

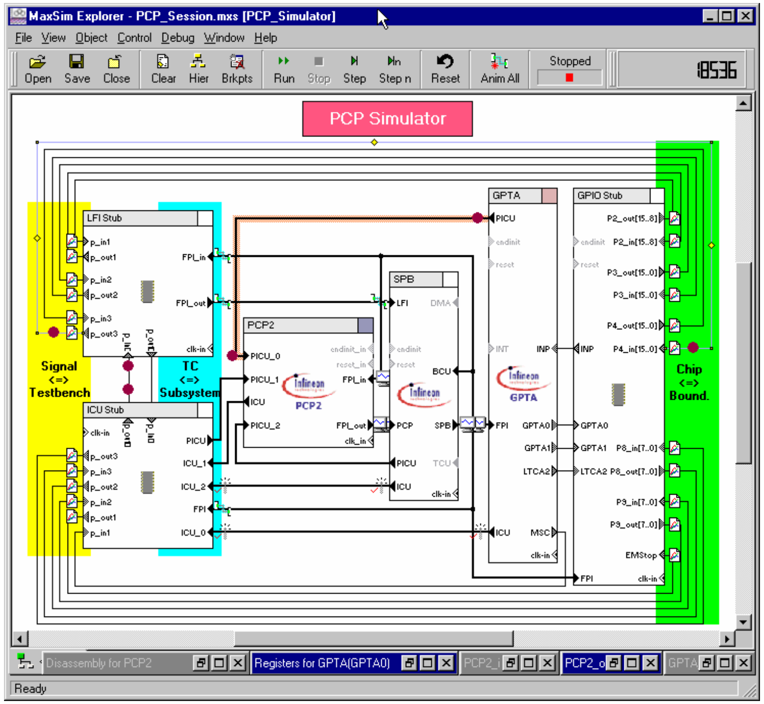


- Key themes:
  - Time to product
  - Right first time
  - Reliability
  - Design reuse
- Automotive
  - System integration
  - Network modeling
  - Benchmarking

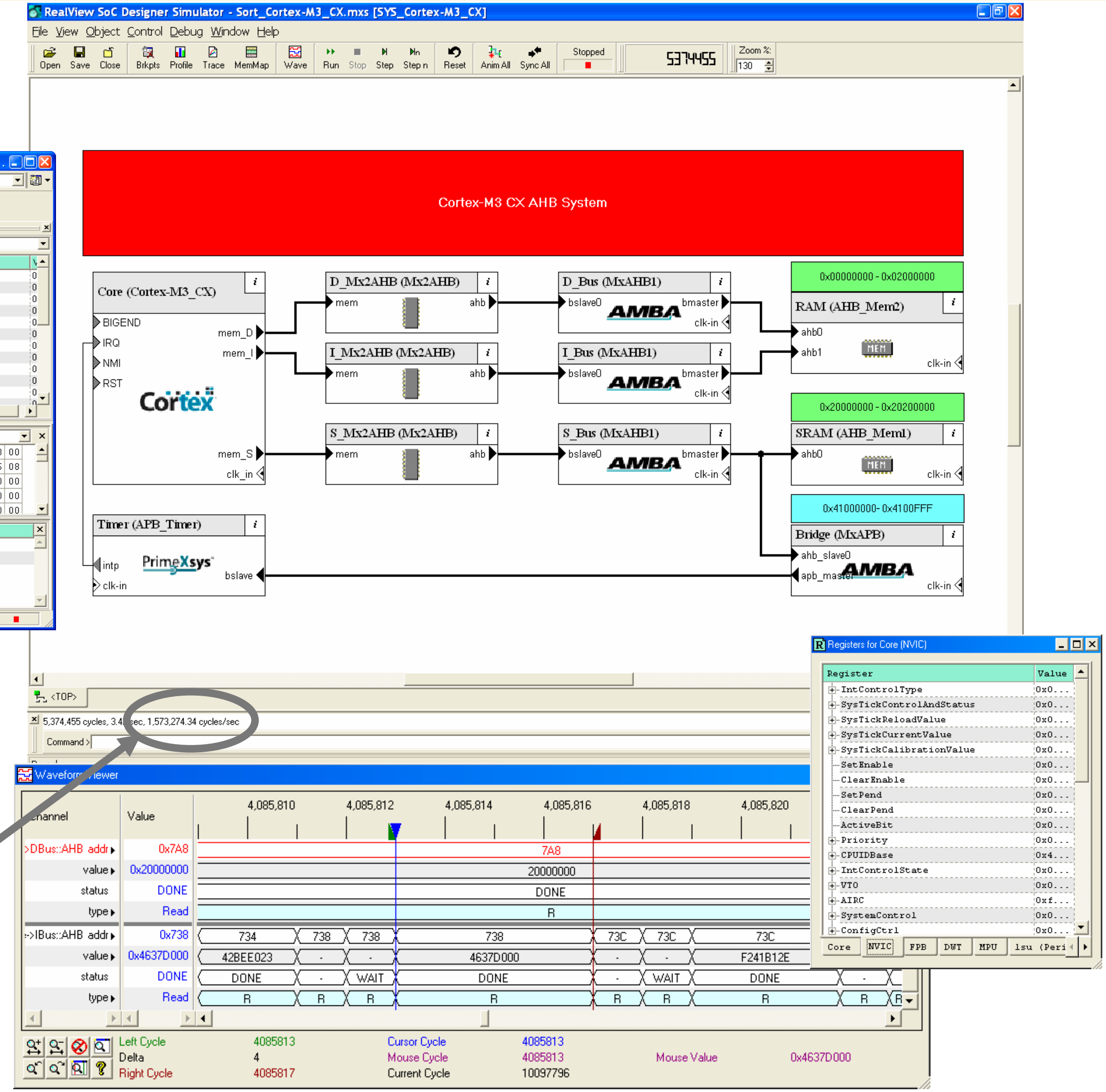
Virtual Prototypes

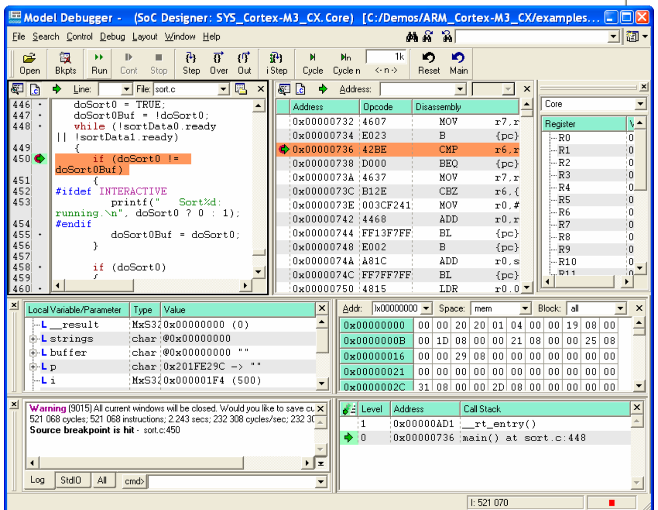
- Allow detailed non-intrusive visibility
  - Exact reproduction of tests
  - Early testing
  - Focused fault diagnostics
- Virtual prototype from IDM for sign off of chip specification

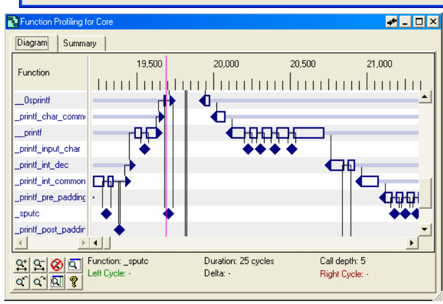




SoC Designer Cortex-M3 AHB System

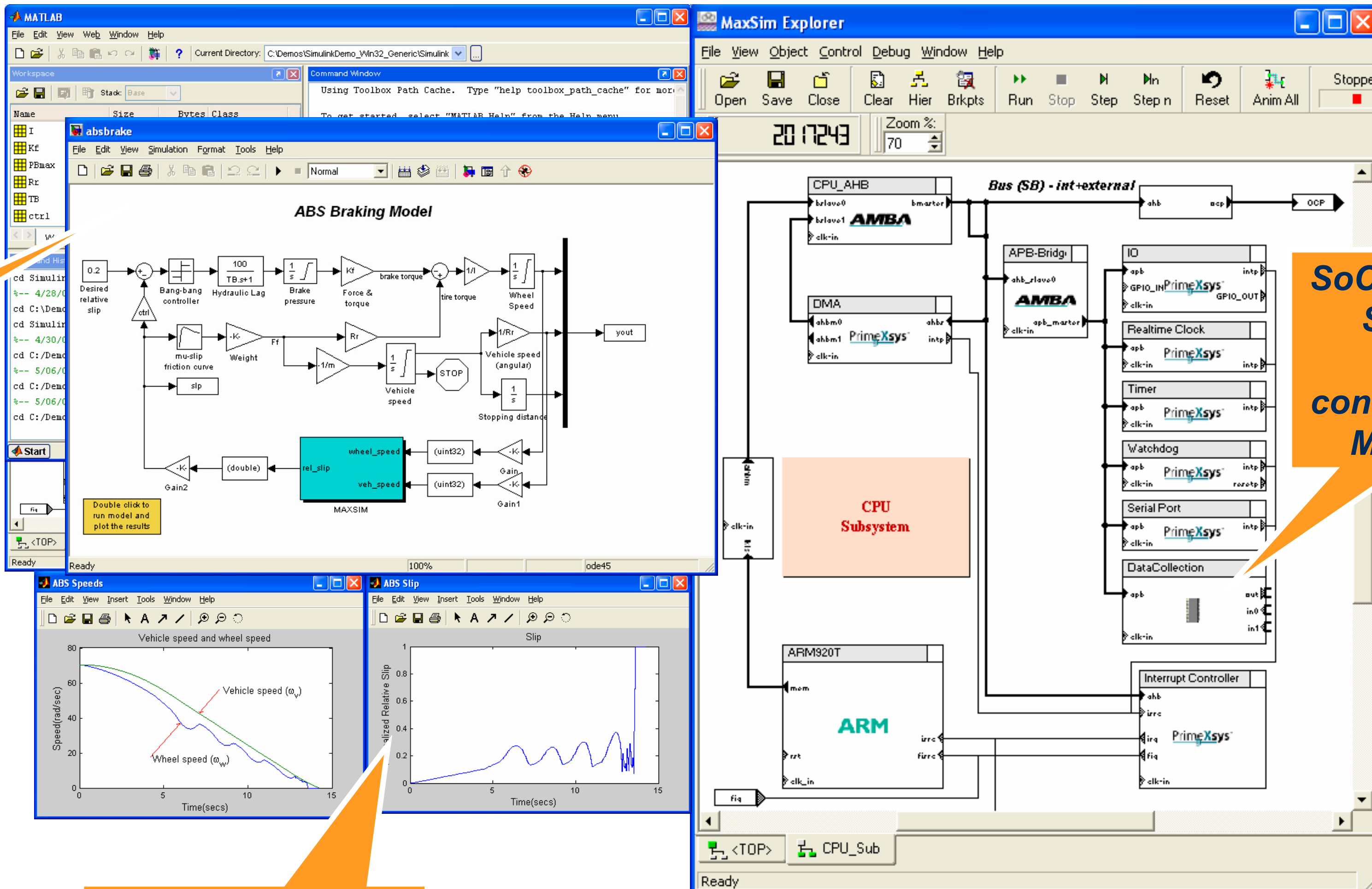






- High Performance: 1.573Mio Cycles/s on a 2GHz Laptop

Co-simulation with Algorithm Tools



MATLAB SimuLink

SoC Designer System with connection to MATLAB

Simulation results

ARM in Automotive

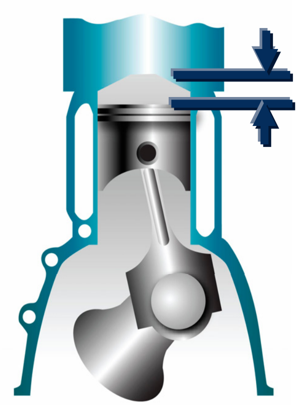
- 65% of worldwide chassis control/ABS
- 50% Design wins of worldwide airbag

Public ARM Automotive Partners

- Autosar OS on Cortex™-M3 processor
  - Reduce complexity and reuse software modules
  - Demonstrated by 3soft at 2006 Embedded World Show
- ARM11™ MPCore™ processor for next generation
  - Navigation and car multimedia applications

ARM Cortex Processors

- Standard architecture
- Tuned for application needs
- Software portability & reuse



- Thumb®-2 Compression
  - 32% size reduction when optimised for -Ospace
  - same size as Thumb
- Thumb-2 Performance
  - 26% size reduction when optimised for -Otime
  - same performance as ARM 32-bit code

Cortex-R4 features for Automotive

- Synthesis configurations for safety-critical systems
  - Parity & ECC support in L1 Cache subsystems
  - External I/F for CPU state for detection of soft and hard errors

	Cortex R4	Benefits
Memory Protection Unit (MPU)	Yes (12 regions)	Sandbox tasks for Reliability
Cache parity & ECC support	Configurable	System reliability
Parallel core logic interface support	Configurable	System reliability and recovery

Cortex-R4: Performance & RAM Access

ARM946E-S	Cortex-R4
At 300MHz	At 400MHz
Required RAM Access Time = 1.3ns	Required RAM Access Time = 2.5ns

ARM Artisan memory compiler data (16Kbits)

Example shows: 53% less power, 35% less area, easier timing closure

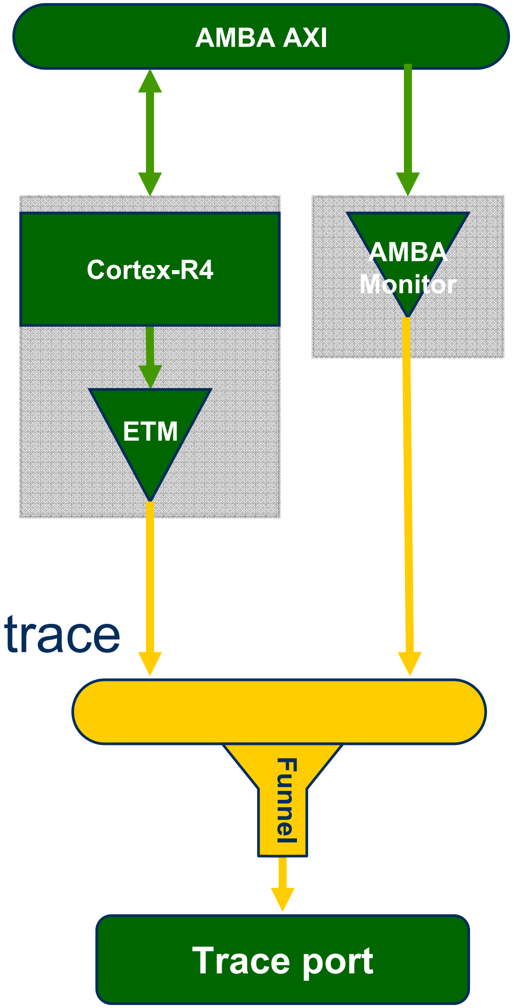
CLN90G 512x32	Advantage RAM compiler	Metro RAM compiler
Access Time	1.23 ns	1.5 ns
Dynamic Power	0.028 mW/MHz	0.013 mW/MHz
Area	0.048 mm^2	0.031 mm^2

Design Options:

- Metro™ RAM for cost effective, low power implementation
- Advantage™ RAM for high performance, robust memory design

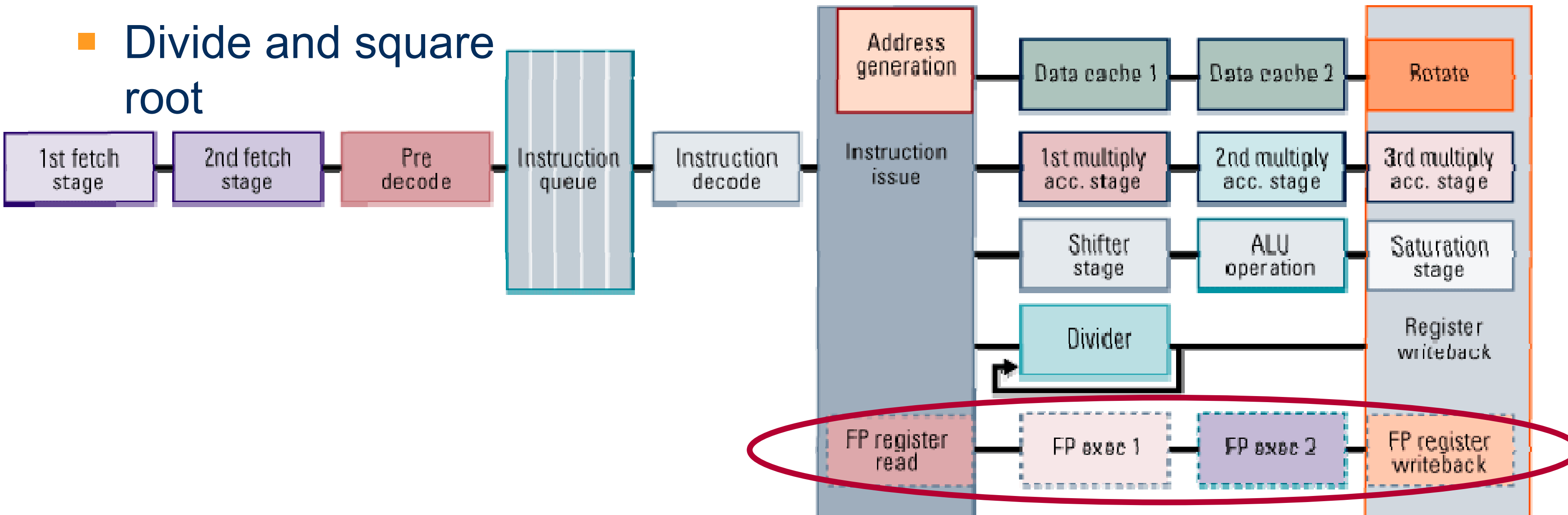
Cortex-R4:CoreSight™ Enhanced Debug

- Trace port speed decoupled from the core
  - Optimise frequency for trace capture
- On-chip trace buffer technology
  - Removes the need for external trace port
  - Use with trace port for best of both worlds
- Higher compression of trace information than ETM7™ and ETM9™
  - Typically 700% better instruction and 25% data trace
  - Either fewer trace pins for instruction trace
    - 4 pins at half core speed
  - Or smaller trace buffers
- Optional system bus monitoring unit
- Trace funnel mixes trace from multiple sources



Serval-EF with Integrated FPU

- Drive train, chassis control and sensor monitor applications
- Dual-issue of FP operations with integer pipeline
- Single precision at near integer level performance
- Hardware support for
  - Single and Double Precision (IEEE 754-compliant)
  - Subnormals, and ‘run fast’ mode
  - Divide and square root

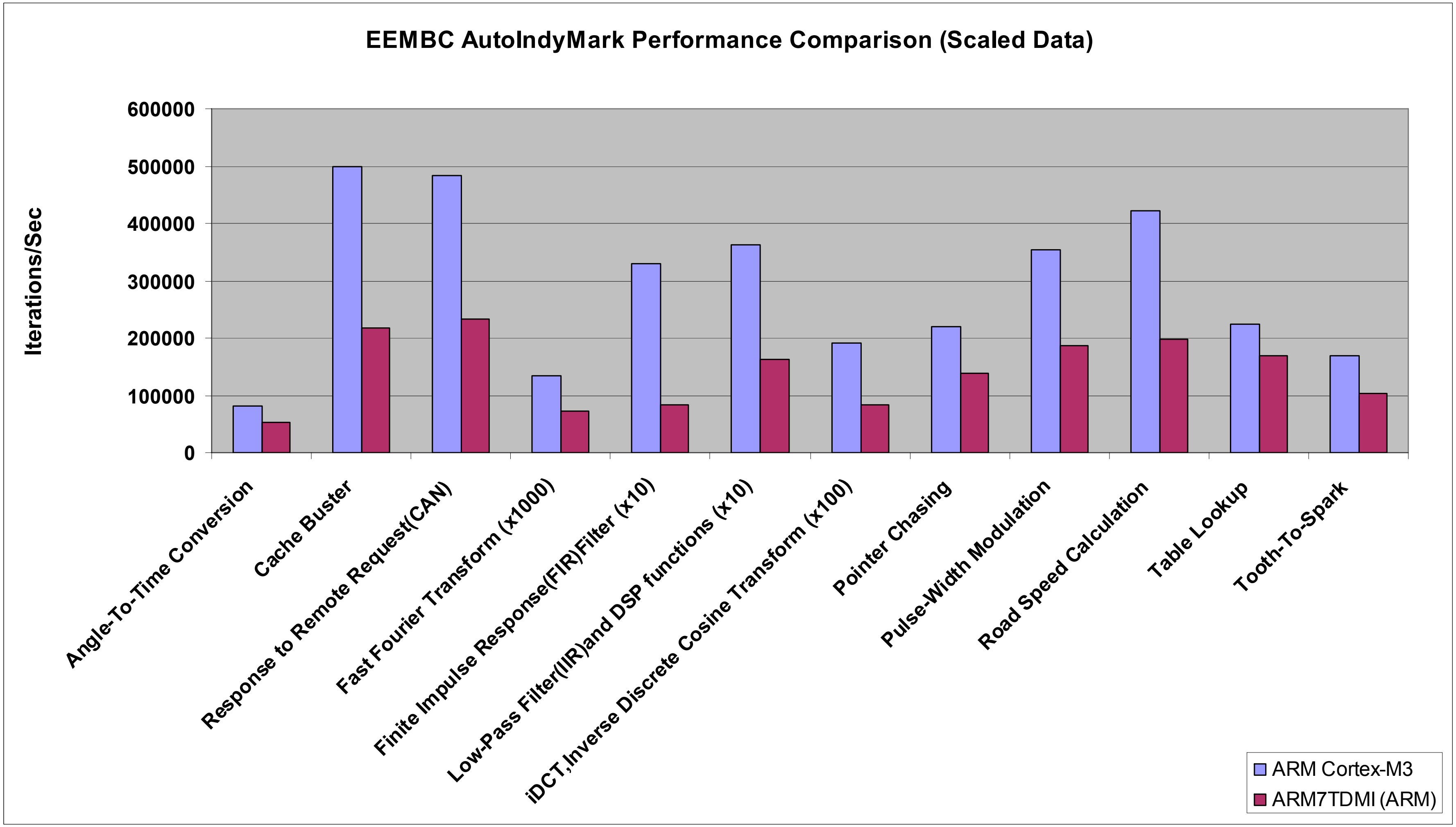


Cortex-M3 for Automotive

- Example applications:
  - Chassis systems; dashboard; ABS; powertrain

	Cortex-M3	Benefits
Optional Memory Protection Unit (MPU)	Yes (12 regions)	Sandbox tasks for reliability
Code size	Thumb-2 16-bit compression	Saves costs & space for more features
Execution from Flash memory	Thumb-2 32-bit performance	32-bit performance with 16-bit costs
Software code update	Flash patch support	Easy software maintenance

RVCT Automotive Benchmarks



Performance comparisons undertaken using standard “out-of-the-box” EEMBC test routines and are for guidance only.
All tests were completed within ARM Ltd and have not been independently validated.

Tests performed between pre-alpha version of ARM Cortex-M3 (Thumb-2 ISA) and ARM7TDMI (ARM ISA)

Tools for Automotive

- RealView® DEVELOP for Automotive
  - Collaboration with 3Soft for AutorSAR
  - High performance compilation, EEMBC automotive benchmarks for Thumb-2 technology
  - Early benchmarking on FPGA, models available from day 1
  - Supplier dependable – well established toolchain for ARM (10+ years)
  - Long term compilation support available
- Cortex-R4, Cortex-M3 trace functionality via ETM support available

ARM and Automotive

- ESL and modeling
  - Co-design with RealView SoC Designer
- Cortex-M3 and Cortex-R4 processors
  - Thumb-2 code compression with performance
  - Reliability with fine grain memory protection
- Autosar OS on Cortex-M3 processor
- ARM KEIL tools integration
  - with key control, UML modeling solutions

Jaguar

Honda Life

VW Golf, Touran

Acura, Fiat, Alfa Romeo, Lancia